



PROBABILITY LECTURE



PROBABILITY



Probability: The likelihood that something will occur.

- It's used in many areas of study, including mathematics, science, business finance, and, in our case, gambling.
- In these disciplines, probability is the number of times something will happen out of the total number of chances of it happening.
- It's commonly expressed as either a fraction ($1/3$) or a percentage (33.3%).
- For example, if I say there is a 60% chance of rain today, I am saying the probability of it raining today is 60 out of 100, which is the same as $60/100$, $6/10$, $3/5$, or 60%.





PROBABILITY OF BEING DEALT POCKET ACES

- Since there are 4 Aces in a deck ($A\clubsuit A\heartsuit A\spadesuit A\diamondsuit$), the probability of being dealt one Ace is 4 in 52.
- Once we're dealt one Ace, there are now only 3 Aces left in the deck of 51 remaining cards; therefore, the odds of our second card also being an ace is 3 in 51.
- We combine these two probabilities together to get a 0.452% chance of being dealt pocket Aces:



$$(4/52) \times (3/51) = 0.452\% \text{ Probability}$$

- This probability holds true for any poker pair if you are asking the probability of being dealt a specific pocket pair before the hand is dealt by the dealer.



ODDS



- Odds are another way of expressing probability.
- Odds are commonly expressed as ratios, such as 2:1, which is stated as 2-to-1.
- In poker, odds play two major roles, the first being our odds or probability of making made hands, such as flopping a set or better, which we determined to be 7.5:1.
- The second is pot odds, which are the odds we offer our opponents when we bet and the odds we are accepting when we call a bet.



CONVERTING ODDS TO FRACTIONS

- For lots of people, including myself, interpreting odds as a ratio can be a bit cryptic.
- I have found that converting and viewing odds as a percentage is much more intuitive.
- The equation on the slide shows a very simple mathematical relationship between a ratio (m:n) and its counterpart fraction, which can be reduced into a percentage.

$$m:n = \frac{n}{m+n}$$

Fraction

Ratio



CONVERTING ODDS TO FRACTIONS



In this mathematical relationship, m is the reward, and n is the risk, and the relationship becomes:

$$\text{Reward: Risk} = \frac{\text{Risk}}{\text{Reward} + \text{Risk}}$$

- m = Reward
- n = Risk
- $m + n$ = Reward + Risk
- **Fraction** = $n / (m + n)$ = Risk / (Reward + Risk)



CONVERTING ODDS EXAMPLES

Converting 2:1 Drawing Odds

- Given 2:1 $\rightarrow$ m:n, where m = 2 & n = 1
- Percentage = $n / (m + n)$
- Percentage = $1 / (2+1) = 1/3$
- 1/3 then converts to 33.3% odds
- So 2:1 drawing odds is equal to **33.3%** drawing odds

Converting 3:1 Pot Odds

- Given 3:1 $\rightarrow$ m:n, where m = 3 & n = 1
- Percentage = $n / (m + n)$
- Percentage = $1 / (3+1) = 1/4$
- 1/4 then converts to 25% odds
- So 3:1 pot odds is equal to **25%** pot odds



ODDS-TO-PERCENTAGE TABLE

- While it's not very difficult to convert odds to percentages once you get the hang of it, you'll often only have seconds to act at the poker table.
- Time is increasingly scarce if you're playing online poker.
- So I've created a simple reference "odds-to-percentage" table that you can use.

Odds Ratios	Odds Percentages
1:1	50%
2:1	33%
3:1	25%
4:1	20%
5:1	16.7%
6:1	14.3%
7:1	12.5%